習題 1-23

(1) $\mathcal{F}[f(x)\, e^{\pm i\omega_0 x}] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} f(x)\, e^{\pm i\omega_0 x}\, e^{-i\omega x}\, dx = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} f(x)\, e^{-i(\omega\mp\omega_0)x}\, dx = \hat{f}(\omega\mp\omega_0)$

(2) $\mathcal{F}[f(kx)] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} f(kx)\, e^{-i\omega x}\, dx = \cancel{\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} f(y)\, e^{-i\omega(\frac{y}{k})}\, d(\frac{y}{k}) = \frac{1}{k\sqrt{2\pi}}\int_{-\infty}^{+\infty} f(x)\, e^{-i(\frac{\omega}{k})x}\, dx}$

當 $k>0$ 時 $= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} f(x)\, e^{-i\omega(\frac{x}{k})}\, d(\frac{x}{k}) = \frac{1}{k}\cdot\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} f(x)\, e^{-i(\frac{\omega}{k})x}\, dx = \frac{1}{k}\hat{f}(\frac{\omega}{k})$

當 $k<0$ 時 $= \frac{1}{\sqrt{2\pi}}\int_{+\infty}^{-\infty} f(x)\, e^{-i\omega(\frac{x}{k})}\, d(\frac{x}{k}) = \frac{1}{-k}\cdot\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} f(x)\, e^{-i(\frac{\omega}{k})x}\, dx = \frac{1}{-k}\hat{f}(\frac{\omega}{k})$

$\Rightarrow \mathcal{F}[f(kx)] = \frac{1}{|k|}\hat{f}(\frac{\omega}{k})$

(3) $i\frac{d}{d\omega}\hat{f}(\omega) = i\cdot\frac{d}{d\omega}\left[\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} f(x)\, e^{-i\omega x}\, dx\right] = i\cdot\int_{-\infty}^{+\infty} f(x)\cdot(-ix)\, e^{-i\omega x}\, dx$

$= \int_{-\infty}^{+\infty} x f(x)\, e^{-i\omega x}\, dx = \mathcal{F}[x f(x)]$

(4) $\mathcal{F}[\int^{x} f(\xi)\, d\xi] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}\left(\int^{x} f(\xi)\, d\xi\right) e^{-i\omega x}\, dx = \frac{i}{\omega\sqrt{2\pi}}\left[\left(\int^{x} f(\xi)\, d\xi\right) e^{-i\omega x}\Big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} f(x)\, e^{-i\omega x}\, dx\right]$

$= -\frac{i}{\omega}\cdot\mathcal{F}[f(x)] = \frac{1}{i\omega}\hat{f}(\omega)$

(5) $\mathcal{F}[f_1(x) * f_2(x)] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}\left(\int_{-\infty}^{+\infty} f_1(\xi)\, f_2(x-\xi)\, d\xi\right) e^{-i\omega x}\, dx$

$= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}\left(\int_{-\infty}^{+\infty} f_2(x-\xi)\, e^{-i\omega x}\, dx\right) f_1(\xi)\, d\xi$

$= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}\left(\int_{-\infty}^{+\infty} f_2(x)\, e^{-i\omega x}\cdot e^{-i\omega\xi}\right) f_1(\xi)\, d\xi$

$= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} \sqrt{2\pi}\,\mathcal{F}[f_2(x)]\cdot f_1(\xi)\cdot e^{-i\omega\xi}\, d\xi$

$= \frac{2\pi}{\sqrt{2\pi}}\mathcal{F}[f_1(x)]\,\mathcal{F}[f_2(x)] = \sqrt{2\pi}\,\mathcal{F}[f_1(x)]\,\mathcal{F}[f_2(x)]$

(三)

习题 1.24

(1) $\mathcal{F}[f(x)]=\frac{1}{\sqrt{2\pi}}\int_{-\pi}^{\pi}\sin x\, e^{-i\omega x}dx=\frac{1}{2\sqrt{2\pi}\,i}\int_{-\pi}^{\pi}[e^{i(-\omega+1)x}-e^{i(-\omega-1)x}]dx$

$=\frac{1}{2\sqrt{2\pi}}\cdot\left[\frac{1}{\omega-1}\left(e^{-i(\omega-1)x}\Big|_{-\pi}^{\pi}\right)-\frac{1}{\omega+1}\left(e^{-i(\omega+1)x}\Big|_{-\pi}^{\pi}\right)\right]$

$=\frac{1}{2\sqrt{2\pi}}\left[\frac{1}{\omega-1}\left(e^{i\omega\pi}-e^{-i\omega\pi}\right)-\frac{1}{\omega+1}\left(e^{i\omega\pi}-e^{-i\omega\pi}\right)\right]$

$=\frac{i}{\sqrt{2\pi}}\sin\omega\pi\cdot\frac{2}{\omega^2-1}\quad(|\omega|\neq 1)$

$\hat{f}(1)=\frac{1}{\sqrt{2\pi}}\int_{-\pi}^{\pi}\sin x\, e^{-ix}dx=-\frac{i}{2\sqrt{2\pi}}\int_{-\pi}^{\pi}(1-\cos 2x)\,dx=-\frac{\sqrt{\pi}}{2\sqrt{2}}i$

$\hat{f}(-1)=\frac{1}{\sqrt{2\pi}}\int_{-\pi}^{\pi}\sin x\, e^{ix}dx=\frac{\sqrt{\pi}}{2\sqrt{2}}i$

(2) $\mathcal{F}[f(x)]=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}\frac{\cos x}{x^2+a^2}e^{-i\omega x}dx=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}\frac{e^{i(1-\omega)x}+e^{-i(1+\omega)x}}{x^2+a^2}dx=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}\frac{e^{i(-\omega+1)x}+e^{i(-\omega-1)x}}{x^2+a^2}dx$

当 $-\omega+1>0$ 时，$\int_{-\infty}^{+\infty}\frac{e^{i(-\omega+1)x}}{x^2+a^2}=2\pi i\left(\frac{e^{-a(-\omega+1)}}{2ai}\right)=\frac{\pi}{a}e^{-a(-\omega+1)}$

当 $-\omega+1<0$ 时，$\int_{-\infty}^{+\infty}\frac{e^{i(-\omega+1)x}}{x^2+a^2}=-2\pi i\cdot\frac{e^{a(-\omega+1)}}{-2ai}=\frac{\pi}{a}e^{a(-\omega+1)}$

$\Rightarrow\int_{-\infty}^{+\infty}\frac{e^{i(-\omega+1)x}}{x^2+a^2}dx=\frac{\pi}{a}e^{-a|-\omega+1|},\quad\omega\neq 1$

当 $-\omega+1=0$ 时，$\int_{-\infty}^{+\infty}\frac{1}{x^2+a^2}dx=\frac{a}{a^2}\int_{-\infty}^{+\infty}\frac{1}{(\frac{x}{a})^2+1}d\left(\frac{x}{a}\right)=\frac{1}{a}\arctan\left(\frac{x}{a}\right)\Big|_{-\infty}^{+\infty}=\frac{\pi}{a}$

$\Rightarrow\int_{-\infty}^{+\infty}\frac{e^{i(-\omega+1)x}}{x^2+a^2}dx=\frac{\pi}{a}e^{-a|\omega+1|}$, $\int_{-\infty}^{+\infty}\frac{e^{i(-\omega-1)x}}{x^2+a^2}dx=\frac{\pi}{a}e^{-a|\omega+1|}$

$\Rightarrow\mathcal{F}[f(\omega)]=\frac{\pi}{a}\left(e^{-a|1-\omega|}+e^{-a|1+\omega|}\right)\cdot\frac{1}{\sqrt{2\pi}}=\frac{1}{a}\sqrt{\frac{\pi}{2}}\left(e^{-a|1-\omega|}+e^{-a|1+\omega|}\right)$

(3) $\mathcal{F}[f(x)]=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}\frac{e^{-i\omega x}}{x^2-1}dx=\frac{1}{\sqrt{2\pi}}\cdot\pi i\left(\frac{e^{i|\omega|}}{-2}+\frac{e^{-i|\omega|}}{2}\right)=\sqrt{\frac{\pi}{2}}\cdot\sin|\omega|$

(4) $\mathcal{F}[f(\omega)]=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}\frac{e^{-i\omega x}}{(x+ai)(x-ai)(x+b)}dx=\begin{cases}2\pi i\cdot\left(\frac{e^{a\omega}}{2ai}\right)\cdot\frac{1}{(b+ai)}+\pi i\,\frac{e^{i\omega b}}{a^2+b^2}, & \omega\leq 0\\ +2\pi i\cdot\frac{e^{-a\omega}}{+2ai}\cdot\frac{1}{(-ai+b)}-\pi i\cdot\frac{e^{i\omega b}}{a^2+b^2}, & \omega>0\end{cases}$

$=\frac{\pi}{a}e^{-a|\omega|}\frac{1}{(\operatorname{sgn}(-\omega)ai+b)}+\operatorname{sgn}(-\omega)\cdot\pi i\cdot\frac{e^{i\omega b}}{a^2+b^2}$

习题 1.25

$\mathcal{F}[xe^{-x^2}] = i\cdot\frac{d}{d\omega}\mathcal{F}[e^{-x^2}]$，$\mathcal{F}[e^{-x^2}] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} e^{-x^2}e^{-i\omega x}dx = \frac{i}{\omega\sqrt{2\pi}}\left[e^{-x^2-i\omega x}\Big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} e^{-i\omega x}\cdot e^{-x^2}(-2x)dx\right]$

$\cancel{= \frac{2i}{\omega}\mathcal{F}[xe^{-x^2}] = -\frac{2}{\omega}\frac{d}{d\omega}\mathcal{F}[e^{-x^2}]}$

$= \frac{2i}{\omega}\mathcal{F}[xe^{-x^2}] = -\frac{2}{\omega}\frac{d}{d\omega}\mathcal{F}[e^{-x^2}]$

$\hat{f}(\omega) = -\frac{2}{\omega}\frac{d}{d\omega}\hat{f}(\omega) \Leftrightarrow -\omega d\omega = 2\frac{d\hat{f}(\omega)}{\hat{f}(\omega)}$

$\Rightarrow -\frac{1}{4}\omega^2 = \ln\hat{f}(\omega) + C$

$\Rightarrow \hat{f}(\omega) = Ce^{-\frac{1}{4}\omega^2}$，$\hat{f}(0) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}e^{-x^2}dx = \frac{1}{\sqrt{2}}$

$\Rightarrow \hat{f}(\omega) = \frac{1}{\sqrt{2}}e^{-\frac{1}{4}\omega^2} = \mathcal{F}[e^{-x^2}]$

$\Rightarrow \mathcal{F}[xe^{-x^2}] = \frac{i}{\sqrt{2}}\cdot\left(-\frac{1}{2}\omega\right)e^{-\frac{1}{4}\omega^2} = \frac{\omega}{2\sqrt{2}i}e^{-\frac{\omega^2}{4}}$

$\mathcal{F}^{-1}\ xe^{-x^2} = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}\frac{1}{2\sqrt{2}i}\omega e^{-\frac{\omega^2}{4}}e^{i\omega x}d\omega = \frac{1}{4\sqrt{\pi}i}\int_{-\infty}^{+\infty}\omega e^{-\frac{\omega^2}{4}}(\cos\omega x + i\sin\omega x)d\omega$

知 $\omega e^{-\frac{\omega^2}{4}}\cos\omega x$ 関於 ω 是奇函数，$\omega e^{-\frac{\omega^2}{4}}\cos\omega x$ 是偶函数

$\Rightarrow \frac{1}{2}xe^{-x^2} = \frac{1}{4\sqrt{\pi}}\int_0^{+\infty}\omega e^{-\frac{\omega^2}{4}}\sin\omega x\,d\omega \Leftrightarrow \int_0^{+\infty}\omega e^{-\frac{\omega^2}{4}}\sin\omega x = 2\sqrt{\pi}\,xe^{-x^2}$

习题 1.26

$\mathcal{F}[f(x)] = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}f(x)e^{-i\omega x}dx = \frac{1}{\sqrt{2\pi}}\int_{-1}^{1}(1-x^2)e^{-i\omega x}dx = \frac{2}{\sqrt{2\pi}}\int_0^1(1-x^2)\cos\omega x\,dx$

$= \frac{2}{\omega\sqrt{2\pi}}\left[[\sin\omega x(1-x^2)]_0^1 - \int_0^1\sin\omega x\cdot(-2x)dx\right] = \frac{2}{\omega}\sqrt{\frac{2}{\pi}}\int_0^1 x\sin\omega x\,dx$

$= \frac{2}{\omega^2}\sqrt{\frac{2}{\pi}}\left[[-\cos\omega x\cdot x]_0^1 - \int_0^1 -\cos\omega x\,dx\right] = \frac{2}{\omega^2}\sqrt{\frac{2}{\pi}}\left(-\cos\omega + \frac{1}{\omega}\sin\omega\right) = 2\sqrt{\frac{2}{\pi}}\frac{\sin\omega - \omega\cos\omega}{\omega^3}$

$f(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}2\sqrt{\frac{2}{\pi}}\frac{\sin\omega-\omega\cos\omega}{\omega^3}e^{i\omega x}d\omega = \frac{4}{\pi}\int_0^{+\infty}\frac{\sin\omega-\omega\cos\omega}{\omega^3}\cos\omega x\,d\omega$

$\Rightarrow \int_0^{+\infty}\frac{\sin\omega-\omega\cos\omega}{\omega^3}\cos\frac{\omega}{2} = \frac{\pi}{4}f\left(\frac{1}{2}\right) = \frac{\pi}{4}\times\left(1-\left(\frac{1}{2}\right)^2\right) = \frac{3\pi}{16}$

清華大學

习题 1.27

(1) $f_1(x) * f_2(x) = \int_{-\infty}^{+\infty} f_1(\xi) f_2(x-\xi)\, d\xi = \int_{-\infty}^{+\infty} f_1(-\xi) f_2(\xi - x)\, d\xi = \int_{-\infty}^{+\infty} f_1(\xi) f_2(-x-\xi)\, d\xi$

$= f_1(-x) * f_2(-x)$

$= \cancel{\int_{-\infty}^{+\infty} f_1(x-\tau) f_2(}$

(只需考虑 $x \geq 0$) $= \int_{-1+|x|}^{1} d\xi = 2-|x|$ (当 $|x|<2$ 时)

当 $|x|>2$ 时, $\forall \tau \in \mathbb{R}, f_1(\tau) f_2(x-\tau) = 0 \Rightarrow f_1(x) * f_2(x) = 0$.

$\Rightarrow f_1(x) * f_2(x) = \begin{cases} 2-|x|, & |x|<2 \\ 0, & |x| \geq 2 \end{cases}$

(2) $f_1(x) * f_2(x) = \int_{-\infty}^{+\infty} f_2(\xi) f_1(x-\xi)\, d\xi$

$= \int_0^{\frac{\pi}{2}} f_2(\xi) f_1(x-\xi)\, d\xi$

当 $x \leq 0$ 时, ~~故~~ $f_1(x) * f_2(x) = 0$

当 $x \in (0, \frac{\pi}{2}]$ 时, $f_1(x) * f_2(x) = \int_0^{x} \sin\xi \, e^{-x} e^{\xi}\, d\xi = \cancel{e^{-x} \int_0^x \sin\xi \, d(e^{\xi})}$

$\cancel{= e^{-x}\left[e^{x} \sin x -\right.}$

$= e^{-x} \operatorname{Im}\left[\int_0^x e^{(1+i)\xi} d\xi\right]$

$= e^{-x} \operatorname{Im}\left(\frac{1}{1+i}\left(e^{(1+i)x} - 1\right)\right)$

$= e^{-x} \operatorname{Im}\left(\frac{1}{\sqrt{2}} e^{x} \cdot e^{i(x-\frac{\pi}{4})} - \frac{1}{\sqrt{2}}(1-i)\right)$

$= e^{-x}\left[\frac{e^{x}}{\sqrt{2}} \sin(x-\frac{\pi}{4}) + \frac{1}{\sqrt{2}}\right] = \frac{\sin(x-\frac{\pi}{4})}{\sqrt{2}} + \frac{e^{-x}}{\sqrt{2}}$

当 $x > \frac{\pi}{2}$ 时, $f_1(x) * f_2(x) = e^{-x} \operatorname{Im}\left[\int_0^{\frac{\pi}{2}} e^{(1+i)\xi} d\xi\right] = e^{-x}\left[\frac{e^{\frac{\pi}{2}}}{2} + \frac{1}{\sqrt{2}}\right]$